

Monte Carlo Simulation - Project

Emil Beck Aagaard, Kamille Møller Frost

October 11, 2025

1 Portfolio loss and importance sampling.

Let $V(S)$ be the value of a portfolio value consisting of financial instruments based on m market prices $S \in \mathbb{R}^m$. The goal of this project is to use Monte Carlo to estimate the portfolio loss L over short time periods ΔT . Here, it is common to use the delta-gamma representation, where the loss is written as

$$L := \delta^\top \Delta S + (\Delta S)^\top \Gamma \Delta S,$$

where $\delta \in \mathbb{R}^m$, $\Gamma \in \mathbb{R}^{m \times m}$ and $\Delta S \sim N(0, \Sigma_S)$ for some covariance matrix Σ_S . Sometimes, it is helpful to consider the linear approximation

$$Q := \delta^\top \Delta S.$$

First, assume that $m = 1$, $\delta = 2$, $\Gamma = \Sigma_S = 1$, and $x = 10$. Please always justify your answer.

```
[104]: # setup (only needed once, this cell is hidden in export)
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from scipy.stats import norm, multivariate_normal, ncx2
```

1.1 Questions

Exercise a) Write a program to estimate the exceedance probabilities $\mathbb{P}(L > x)$ by crude Monte Carlo simulation.

For a better comparison of our results, we first derive the analytical probability, for the simple case $m=1$, $\delta=2$ and $\Gamma = \Sigma_S = 1$. Using these parameters we can write:

$$\mathbb{P}(L > x) = \mathbb{P}(2Z + Z^2 > x) \quad Z \sim N(0, 1)$$

Completing the square we get $2Z + Z^2 = (Z + 1)^2 - 1 = Y^2 - 1$ where $Y \sim X_{nc}^2(\nu = 1, \lambda = 1)$, and where $X_{nc}^2(\nu, \lambda)$ denotes a non-central chi-square distribution with ν degrees of freedom and non-centrality parameter λ . In particular the analytical probability is can be calculated by:

$$\mathbb{P}(L > x) = 1 - \mathbb{P}(Y^2 - 1 \leq x) = 1 - \mathbb{P}(Y^2 \leq x + 1)$$

This is determined to be

```
[105]: x = 10
analytical_prop = 1-ncx2.cdf(x+1, df=1, nc=1)
print(f"Analytical probability P(L > {x}): {analytical_prop:.6f}")
```

Analytical probability P(L > 10): 0.010270

We now implement a crude Monte Carlo simulation, of this probability. We first define:

```
[106]: np.random.seed(2) # for reproducibility
# assumptions
m = 1
delta = 2
gamma = 1
sigma = np.array([[1]]) # covariance matrix

def loss(m, delta, delta_S, gamma):
    """ Compute the loss L given delta, gamma, sigma and delta_S, note that all
    ↪ parameters should be of dimension m."""
    if m == 1:
        Q = delta*delta_S
        L = Q + (delta_S * gamma) * delta_S
    else:
        Q = delta_S@delta
        L = Q + np.sum((delta_S @ gamma) * delta_S, axis=1)
    return L
```

Crude Monte Carlo is then done by finding N realizations of $\Delta S \sim N(0,1)$. For each realization we then determine the *empirical* loss $\tilde{L}$ and weather or not $\tilde{L} < x$. In particular the Law of large numbers then gives that

$$\frac{\mathbf{1}_{L_1 < x} + \dots + \mathbf{1}_{L_N < x}}{N} \xrightarrow{N \rightarrow \infty} \mathbb{E}[\mathbf{1}_{L_1 < x}] = \mathbb{P}(L_1 < x) = 0.010270$$

In other words, crude Monte Carlo will converge towards the analytical probability when we increase our sample size parameter N .

For $N = 10^6$ we find:

```
[107]: N = 10**6 # number of simulations

def crude_monte_carlo(N, m, delta, gamma, sigma, x, loss_function):
    """ Crude Monte Carlo simulation of the probability P(L > x), note that all
    ↪ parameters should be of dimension m."""

    delta_S = np.random.multivariate_normal(np.zeros(m), sigma, size=N)
    L = loss_function(m, delta, delta_S, gamma)
    return np.mean(L > x), L
```

```
P_crude, L = crude_monte_carlo(N, m, delta, gamma, sigma, x, loss)
print(f"Crude Monte Carlo estimate of P(L > {x}): {P_crude:.6f}")
print(f"Difference to analytical: {abs(P_crude - analytical_prop):.6f}")
```

Crude Monte Carlo estimate of $P(L > 10)$: 0.010188

Difference to analytical: 0.000082

Which is already a quite small deviation, however for smaller N we find:

```
[108]: N = 100
P_crude_small, L_small = crude_monte_carlo(N, m, delta, gamma, sigma, x, loss)
print(
    f"Crude Monte Carlo estimate of P(L > {x}) with N={N}: {P_crude_small:.6f}")
print(f"Difference to analytical: {abs(P_crude_small - analytical_prop):.6f}")
```

Crude Monte Carlo estimate of $P(L > 10)$ with $N=100$: 0.000000

Difference to analytical: 0.010270

By the above, it is clear that our crude monte carlo estimate very much depends on the value of N .

Exercise b) Use control variables method to reduce the variance of the estimator from part (a). Discuss several possibilities and argue for your final choice. Would you recommend to use these methods given potential longer compute time?

In our Monte Carlo simulation, we want to determine $\mathbb{P}(L < x) = \mathbb{E}[\mathbf{1}_{L < x}] =: \ell$. We know that this can be done naively by the crude Monte Carlo utilizing the Law of Large numbers. However, if we want a high precision of the estimate, we might need a very high sample size N . Another way to determine the quality of our estimates, compared to the absolute difference in numerical value above, is the mean-squared error. For a given set of samples $Y_1, \dots, Y_n$ from the crude Monte Carlo we have:

$$\mathbb{E} \left[\left(\frac{(Y_1 + \dots + Y_n)}{N} - \mathbb{E}[Y_1] \right)^2 \right] = \mathbb{V} \left[\frac{(Y_1 + \dots + Y_n)}{N} \right] = \frac{1}{N} \mathbb{V}[Y_1].$$

The control variables method is one way to guarantee a variance reduction and thus increase the quality of our estimate of ℓ , at least by the above measure of quality. The main idea is that apart from sampling variables of the form $Y = \mathbf{1}_{L < x}$ as in the crude Monte Carlo, we also calculate Z , where $\mathbb{E}[Z]$ is known, and where Y and Z are positively correlated. We then use $\hat{Y} = Y - (Z - \mathbb{E}[Z])$ as our new variable for the Law of large numbers, indeed:

$$\mathbb{E}[\hat{Y}] = \mathbb{E}[Y] - 0 = \ell$$

In particular, we have shown in the lectures, that for the class of estimators given by

$$\hat{\ell} := \frac{1}{N} \sum_{i \leq N} [Y - \beta(Z_i - \ell_Z)]$$

there exists an optimal β given by

$$\beta_{\text{opt}} = \frac{\text{Cov}[Y, Z]}{\mathbb{V}(Z)}$$

and the minimum value of the variance is

$$\frac{1}{N} \mathbb{V}(Y) (1 - \text{Corr}[Y, Z]^2)$$

By the above, the a variance reduction is then guarenteed since Y and Z was assumed to be positively correlated.

To implement this method we will use the proposed estimator Q stated in the project description, and consider $\mathbf{1}_{Q > (x/2)}$

```
[109]: def cv_estimator(Z, x, delta, m, analytical_mean):
    if m == 1:
        H = delta*Z
    else:
        H = Z@delta

    return H > x/2, np.mean(H > x/2)
```

In particular, for the given parameters, we have $\mathbb{E}[\mathbf{1}_{Q > (x/2)}] = \mathbb{P}[2\Delta S \geq x/2] = 1 - \phi(x/4)$, where $\Delta S \sim N(0, 1)$

We can then implement Control variance for this estimator:

```
[110]: N = 10**6

def control_variance_monte_carlo(N, m, delta, x, control_variance_estimator,
    analytical_mean):
    """ Control Variance monte carlo of P(L>x), note that all parameters shold
    be of dimension m. """

    Z = np.random.multivariate_normal(np.zeros(m), sigma, N)
    Y = loss(m, delta, Z, gamma) > x
    H, l_H = control_variance_estimator(Z, x, delta, m, analytical_mean)

    model = LinearRegression()
    beta = model.fit(H.reshape(-1, 1), Y).coef_[0]

    print(f"Optimal beta: {beta.item():.4f}")

    Y_hat = Y - beta*(H - l_H)

    return np.mean(Y_hat), Y, Y_hat

P_cv, Y, Y_hat = control_variance_monte_carlo(
    N, m, delta, x, cv_estimator, norm.cdf(x/4))
print(
    f"Control Variance using Q in Monte Carlo estimate of P(L > {x}): {P_cv:.
    6f}")
```

```

print(f"Variance reduction: {np.var(Y)/np.var(Y_hat):.2f} times")

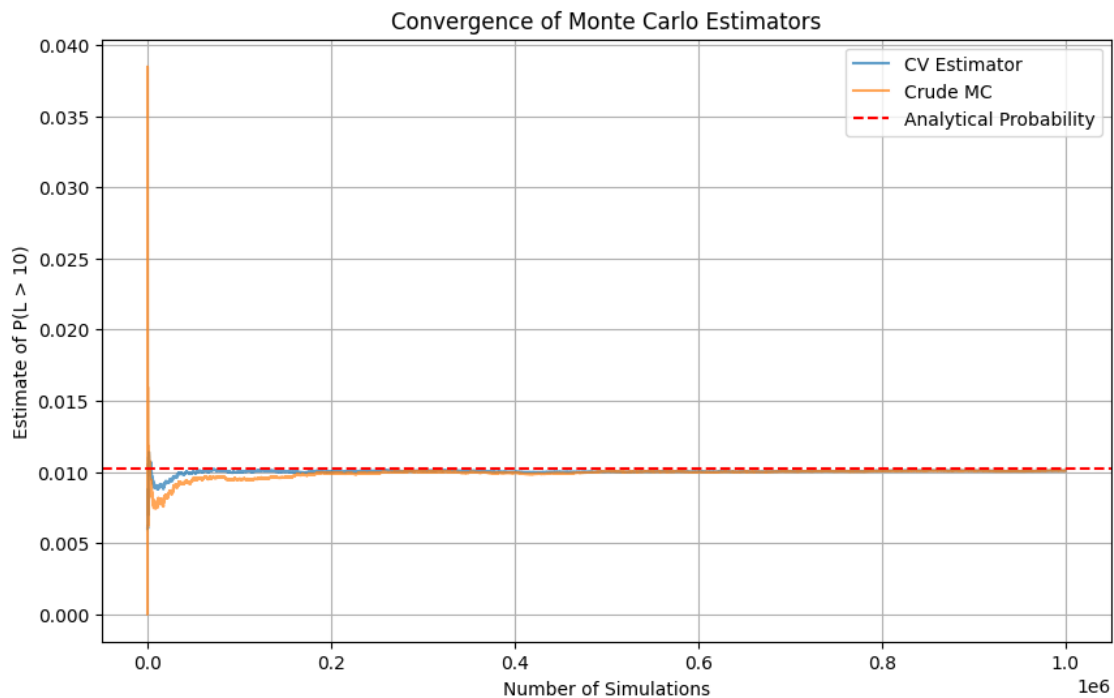
# create a plot of the convergence of the probability estimates
plt.figure(figsize=(10, 6))
plt.plot(np.cumsum(Y_hat)/np.arange(1, N+1),
         label='CV Estimator', alpha=0.7)
plt.plot(np.cumsum(Y)/np.arange(1, N+1), label='Crude MC', alpha=0.7)
plt.axhline(analytical_prop, color='red', linestyle='--',
            label='Analytical Probability')
plt.xlabel('Number of Simulations')
plt.ylabel(f'Estimate of P(L > {x})')
plt.title('Convergence of Monte Carlo Estimators')
plt.legend()
plt.grid()
plt.show()

```

Optimal beta: 0.9959

Control Variance using Q in Monte Carlo estimate of $P(L > 10)$: 0.010085

Variance reduction: 2.46 times



Exercise c) Design an importance-sampling method to reduce the variance of the estimator from part (a). Again, discuss several options and argue for your final choice.

Again, discuss the trade-off between additional compute time and variance reduction.

Exercise d) Create and discuss visualizations for comparing (a), (b) and (c).

Exercise e) Show that for the specific choices of m, δ, Γ and Σ_S , the probability $\mathbb{P}(L > x)$ has a closed-form expression only involving the cdf of the standard normal distribution. Is this still true when choosing another distribution? For the variance-reduction methods in (b), (c), how easy do they generalize when L is replaced by another relevant distribution?

Question 1) Now, assume that $m = 2, \delta = (2, 2)^\top, \Gamma = Id_2, \Sigma_S = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, and $x = 15$. Repeat parts (a) and (b) for these new parameters.

```
[112]: m = 2
delta = np.array([2, 2])
gamma = np.identity(2)
sigma = np.array([[1, 0.5], [0.5, 1]])
x = 15
N = 10**5

P_crude, L = crude_monte_carlo(N, m, delta, gamma, sigma, x, loss)
print(f"Crude Monte Carlo estimate of P(L > {x}): {P_crude:.6f}")

P_cv, Y, Y_hat = control_variance_monte_carlo(
    N, m, delta, x, cv_estimator, 0.5)
print(
    f"Control Variance using Q in Monte Carlo estimate of P(L > {x}): {P_cv:.6f}")
print(f"Variance reduction: {np.var(Y)/np.var(Y_hat):.2f} times")

# create a plot of the convergence of the probability estimates
plt.figure(figsize=(10, 6))
plt.plot(np.cumsum(Y_hat)/np.arange(1, N+1),
         label='CV Estimator', alpha=0.7)
plt.plot(np.cumsum(Y)/np.arange(1, N+1), label='Crude MC', alpha=0.7)

plt.xlabel('Number of Simulations')
plt.ylabel(f'Estimate of P(L > {x})')
plt.title('Convergence of Monte Carlo Estimators')
plt.legend()
plt.grid()
```

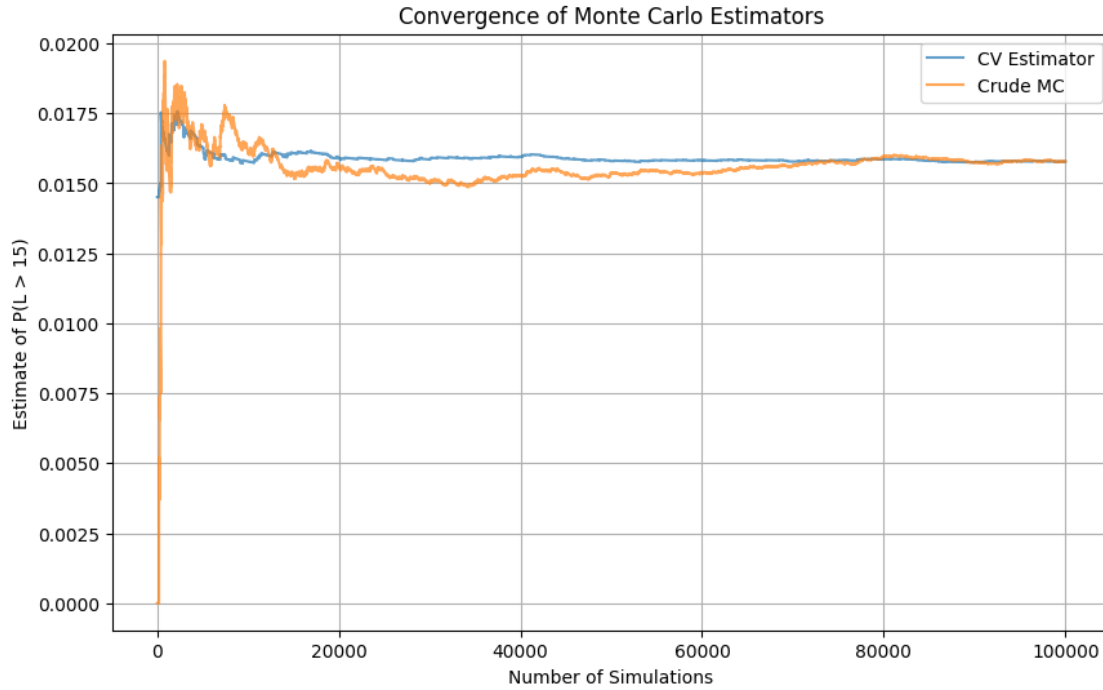
```
plt.show()
```

Crude Monte Carlo estimate of $P(L > 15)$: 0.016270

Optimal beta: 0.9509

Control Variance using Q in Monte Carlo estimate of $P(L > 15)$: 0.015780

Variance reduction: 7.99 times



Exercise f) Now, use the Cholesky decomposition to find a matrix $\tilde{C}$ such that $\Delta S = \tilde{C}Z$ where $Z \sim N(0, I_2)$. Next, find an orthogonal matrix U and a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2)$ such that $\tilde{C}^\top \tilde{C} = U\Lambda U^\top$. Show that the loss L can be expressed as

$$L = b_1 Z_1 + b_2 Z_2 + \lambda_1 Z_1^2 + \lambda_2 Z_2^2$$

where $b = C^\top \delta$ with $C = \tilde{C}U$.

Exercise g) Similarly as in part (c), use $\prod_{i \leq 2} \exp(\theta_i Z_i - \psi^*(\theta_i))$ as an importance-sampling density, where ψ^* is the cumulant generating function of a standard normal distribution. Experiment with different parameters θ_i .

Exercise h) For $(\lambda_1 \vee \lambda_2) \theta < 1/2$ show that

$$\psi_L(\theta) := \log \mathbb{E}[\exp(\theta L)] = \frac{1}{2} \sum_{j \leq 2} \left(\frac{\theta^2 b_j^2}{1 - 2\theta \lambda_j} - \log(1 - 2\theta \lambda_j) \right).$$

You may use without proof that

$$\mathbb{E} \left[\exp \left(\theta (Z_1 + c)^2 \right) \right] = (1 - 2\theta)^{-1/2} \exp(\theta c^2 / (1 - 2\theta))$$

holds for all $c \in \mathbb{R}$ and $\theta < 1/2$.

Exercise i) Use part (h) to develop an importance-sampling algorithm for the estimation of $\mathbb{P}(L > 15)$. Develop a heuristic for choosing θ .

Exercise j) Describe broadly how stratified sampling could be used for simulating L when the random variable Q is used for stratification. For this question, a description of your approach is sufficient; you do not need to implement a simulation program.